Flow Control at Data Link Layer using Sliding Window Protocol.

Aim :-

To write a program to implement flow control at the Data Link Layer using the Sliding window Protocol, and to simulate the flow of frames from one to one to another.

Algorithm.

1. Start the program.
2. Read the total number of frames (N), window size(w) and lost frames (Optional)
3. Initialize . base = 0 and nextSeq = 0.
4. While base < N :
   * Set the Sender window as [base, base + w - 1].
   * Send frames within the window and increase nextSeq.
   * If a frame is lost
     (i) Receiver & Send it and later frames.
     (ii) Sender times out and retransmits from the lost frames.

   Otherwise :
     (i) Receiver Sender ACKs for Received frames.
     (ii) Sender Slides the window.
5) Repeat until all frames are transmitted and acknowledged
6) Stop the program.

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=== SLIDING WINDOW PROTOCOL - FLOW CONTROL SIMULATION (Go-Back-N) ===

Enter total number of frames to send : 8
Enter sender's window size : 3
Enter frame number to simulate as LOST
(0 to 7, enter -1 for no loss): 4

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Sender Window : [Frame 0 - Frame 2]
Node A --> Sending Frame 0
Node A --> Sending Frame 1
Node A --> Sending Frame 2
Node B --> Received Frame 0, sending ACK 0
Node B --> Received Frame 1, sending ACK 1
Node B --> Received Frame 2, sending ACK 2
Node A --> ACK 2 received. Sliding window forward.

-----------------------------------------------
Sender Window : [Frame 3 - Frame 5]
Node A --> Sending Frame 3
Node A --> Sending Frame 4
 (Frame 4 LOST in transit!)
Node A --> Sending Frame 5
Node B --> Frame 4 missing. Discarding out-of-order frames after it.
Node A --> Timeout! Resending window starting from Frame 4.

-----------------------------------------------
Sender Window : [Frame 3 - Frame 5]
Node A --> Sending Frame 4
Node A --> Sending Frame 5
Node B --> Received Frame 3, sending ACK 3
Node B --> Received Frame 4, sending ACK 4
Node B --> Received Frame 5, sending ACK 5
Node A --> ACK 5 received. Sliding window forward.

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Sender Window : [Frame 6 - Frame 7]
Node A --> Sending Frame 6
Node A --> Sending Frame 7
Node B --> Received Frame 6, sending ACK 6
Node B --> Received Frame 7, sending ACK 7
Node A --> ACK 7 received. Sliding window forward.

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All 8 frames successfully transmitted from Node A to Node B.
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Program

```
def simulate (n, w, lost):
    b = ns = 0
    While .b < n:
        e = min (b + w - 1 ; n - 1)
        print ("n. . . . . ")
    Print (f" Sender Window : [Frames {b} - Frames {e}]")
    While .ns < b + w and ns < n!
        Print (f" Node A - - x . Sending Frame {ns}")
    If ns == lost':
        print (f" (Frame {ns}) Lost in transit !)")
    ns += 1
    If lost is not node . and b < = lost < ns.
    Print (f"'Node B - - -> Frame {lost} missing . Discard.")
    Print (f'Node . A - - > Time out {lost}.")
    ns = lost
    Lost = None
    Continue
    for i in range (b, ns):
        Print (f" Node - - -> Received . Frames {i}
                                    Send {i}")
    Print (f" Node A -> A ck {ns - 1} received, Sliding)
    b = ns
```

```
print ("=== Sliding Window Protocol - FlowControl\n")
n = int (input ("Enter total no. of frames to send : "))
w = int ( input ("Enter Sender's window size : "))
While True :
    Try :
        lost = int (input ("Enter frame number [1-3].
            Enter -1 for no loss) : ")).
        if -1 <= lost < n:
            break
        Print ("please enter a value between -1 and [1-3]")
    except ValueError:
        Print ("Invalid input ! Please Enter integer")
Simulate (n, w, None if lost == -1 else lost).
```

Result :-

The program to implement flow control at the Data Link Layer using the sliding Window Protocol was successfully written in python, executed and Verified by simulating the flow of frames between two nodes.